



ROLL NO

PUNJAB PUBLIC SERVICE COMMISSION
COMBINED COMPETITIVE EXAMINATION
FOR RECRUITMENT TO THE POSTS OF
PROVINCIAL MANAGEMENT SERVICE, ETC -2023
CASE NO. 1C2024

SUBJECT: ZOOLOGY (PAPER-I)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

NOTE: **Attempt FIVE Questions in all. Select at least TWO Questions from each Section. Attempt in Urdu or English.**

SECTION-I

- Q.No.1** **a)** Describe different types in skeletons in each class of sponges.
b) Discuss various types of canal system in phylum porifera. **(10+10=20 Marks)**
- Q.No.2** **a)** Describe the life cycle of *Fasciola hepatica*.
b) Describe reproduction and development in *turbellaria*. **(10+10=20 Marks)**
- Q.No.3** **a)** Explain feeding and digestive system of Nematoda.
b) Give an account of reproduction and development in Nematoda. **(10+10=20 Marks)**
- Q.No.4** **a)** Discuss evolutionary relationship of Annelida with other animals.
b) What is metamerism? Discuss its advantages in annelids. **(10+10=20 Marks)**

SECTION-II

- Q.No.5** Describe mechanism of osmoregulation in fresh water and marine fishes. **(20 Marks)**
- Q.No.6** Discuss rise and fall of reptiles during their evolutionary history. **(20 Marks)**
- Q.No.7** Give an account on aerial adaptations and migration of birds. **(20 Marks)**
- Q.No.8** Discuss Amphibians as first terrestrial vertebrates. **(20 Marks)**



ROLL NO

PUNJAB PUBLIC SERVICE COMMISSION
COMBINED COMPETITIVE EXAMINATION
FOR RECRUITMENT TO THE POSTS OF
PROVINCIAL MANAGEMENT SERVICE, ETC -2023
CASE NO. 1C2024

SUBJECT: ZOOLOGY (PAPER-II)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

NOTE: Attempt FIVE Questions in All Selecting ONE from each Section.
Attempt in Urdu or English.

SECTION-A

- Q. No. 1.** a) Describe various models of plasma membrane. Explain the more dynamic model with justification.
b) Elaborate the extra nuclear organelles in eukaryotic cell with their functions.

(10+10=20 Marks)

- Q. No. 2.** a) Elaborate different nuclear and extra nuclear cell organelles along with their functions. Draw the labelled diagrams also.

- b) Differentiate between reversible and irreversible enzymatic inhibition.

(10+10=20 Marks)

SECTION-B

- Q. No. 3.** a) How the osmoregulation contributes in homeostasis? Describe process of osmoregulation in vertebrates.

- b) Write in detail the mechanisms of active membrane potential and resting membrane potential. Support the answer with suitable diagrams.

(10+10=20 Marks)

- Q. No. 4.** a) Describe the mechanism of respiration in animals, write the role of different organs of respiratory system in detail.

- b) What are different organs involved in circulatory system? Write the cardiovascular mechanism comprehensively.

(10+10=20 Marks)

SECTION-C

- Q. No. 5.** a) Differentiate between two principles of Mendel namely the law of segregation and law of independent assortment.

- b) How can pedigree analysis be used to predict genotypes of the future offspring?

(10+10=20 Marks)

- Q. No. 6.** a) Discuss a case of mammalian embryo who has a blood type antigenically different from its mother.

- b) What is epistasis? Explain a pattern of epistatic gene interaction in humans.

(10+10=20 Marks)

SECTION-D

- Q. No. 7.** a) How can genetic drift contribute to evolution? What are the causes of genetic drift?

- b) Why is the natural selection considered a primary mechanism of adaptive evolution?

(10+10=20 Marks)

- Q. No. 8.** a) Which mechanisms provide raw materials for the process of natural selection to act upon?

- b) What is adaptive radiation? Explain a case example of adaptive radiation.

(10+10=20 Marks)

SECTION-E

- Q. No. 9.** a) How agroecosystems differ from natural ecosystems? What are the different types of agroecosystems?
b) Describe a general model of nutrient cycling showing the primary reservoirs of nutrients.

(10+10=20 Marks)

- Q. No. 10.** a) Differentiate between gaseous nutrient cycles and sedimentary cycles.

- b) Briefly describe the major steps in nitrogen cycle.

(10+10=20 Marks)